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Intrinsic neural timescales exhibit different lengths in distinct meditation techniques

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ABSTRACT

Meditation encompasses a range of practices employing diverse induction techniques, each characterized by a distinct attentional focus. In Mantra meditation, for instance, practitioners direct their attention narrowly to a given sentence that is recursively repeated, while other forms of meditation such as Shoonya meditation are induced by a wider attentional focus. Here we aimed to identify the neural underpinnings and correlates associated with this spectrum of distinct attentional foci. To accomplish this, we used EEG data to estimate the brain's intrinsic neural timescales (INTs), that is, its temporal windows of activity, by calculating the Autocorrelation Window (ACW) of the EEG signal. The autocorrelation function measures the similarity of a timeseries with a time-lagged version of itself by correlating the signal with itself on different time lags, consequently providing an estimation of INTs length. Therefore, through using the ACW metric, our objective was to explore whether there is a correspondence between the length of the brain's temporal windows of activity and the width of the attentional scope during various meditation techniques. This was performed on three groups of highly proficient practitioners belonging to different meditation traditions, as well as a meditation-naïve control group. Our results indicated that practices with a wider attentional focus, like Shoonya meditation, exhibit longer ACW durations compared to practices requiring a narrower attentional focus, such as Mantra meditation or body-scanning Vipassana meditation. Together, we demonstrated that distinct meditation techniques with varying widths of attentional foci exhibit unique durations in their brain's INTs. This may suggest that the width of the attentional scope during meditation relates and corresponds to the width of the brain's temporal windows in its neural activity.

Significance Statement: Our research uncovered the neural mechanisms that underpin the attentional foci in various meditation techniques. We revealed that distinct meditation induction techniques, featured by their range of attentional widths, are characterized by varying lengths of intrinsic neural timescales (INTs) within the brain, as measured by the Autocorrelation Window function. This finding may bridge the gap between the width of attentional windows (subjective) and the width of the temporal windows in the brain's neural activity (objective) during different meditation techniques, offering a new understanding of how cognitive and neural processes are related to each other. This work holds significant implications, especially in the context of the increasing use of meditation in mental health and well-being interventions. By elucidating the distinct neural foundations of different meditation techniques, our research aims to pave the way for developing more tailored and effective meditation-based treatments.

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1. Introduction

1.1. Meditation and its different induction methods

Meditation practices have been developed for millennia in several cultural contexts, taking on a variety of unique forms (Komjathy, 2015). They can be conceived as a set of techniques to reduce suffering and foster a more harmonic relationship with the environment (Adventure-Heart and Proeve, 2017; Trautwein et al., 2014; Ray et al., 2021). While this final goal is shared among the different meditation techniques (Cooper et al., 2022; Reddy and Roy, 2018; J.S.K. 2019; Osho, 2012), the methods of inducing such a meditative state can differ substantially (Fox et al., 2016; Raffone et al., 2019; Fucci et al., 2018; Fujino et al., 2018; Lehmann et al., 2012; Awasthi, 2013; Woods et al., 2020; West, 1987; Koshikawa and Ichii, 1996; Lutz et al., 2007). The different induction methods are often characterized by a shift of the practitioner's attentional focus on different objects of choice, ranging from spatially confined objects (such as breath or other bodily sensations) to more expansive spaces (like the room one sits in) (Braboszcz et al., 2017; Jha et al., 2007; Lutz et al., 2008; Raffone and Srinivasan, 2010; Kozhevnikov et al., 2022; Tsai and Chou, 2016; Lippelt et al., 2014; Valentine and Sweet, 1999; Manna et al., 2010; Prakash, 2021; Brandmeyer et al., 2019; Amihai and Kozhevnikov, 2015).

Different induction techniques have their respective attentional width, resulting in a considerable range across all techniques (Lutz et al., 2008; Lippelt et al., 2014). The sample in the present study consisted of three groups of subjects whose contemplative practice was characterized by three different types of induction: (1) Shoonya meditation, (2) Mantra meditation, and (3) Vipassana meditation. (1) Shoonya meditation endeavors to release any specific attentional focus and avoid concentrating on any particular content, thereby promoting an unconstrained free-flow with a broad attentional scope (see Fig. 1a for a visual depiction; Braboszcz et al., 2017, Martínez Vivot et al., 2020; Nash et al., 2013). Conversely, in (2) Mantra meditation the attentional focus is narrowed to one constant and recursive input, i.e., a repeating word, phrase, or sound. The practitioner thus maintains a stable and narrow focus of attention over time on a single stimulus, excluding other possible external and internal stimuli (such as sounds, smells, thoughts, etcetera) (Braboszcz et al., 2017; Álvarez-Pérez et al., 2022; Tseng et al., 2022; Sharma et al., 2022; Harne et al., 2018; Nash et al., 2013; Martínez Vivot et al., 2020; Lippelt et al., 2014; Katyal, 2022; Lutz et al., 2008, Malinowski, 2013). Lastly, in (3) Vipassana meditation, the attentional focus is on somatosensory awareness, directing the attention towards a sequence of bodily stimuli (Braboszcz et al., 2017; Chiesa et al., 2010; Delgado-Pastor et al., 2013; Nash et al., 2013; Zeng et al., 2014; X. 2015; Cahn and Polich, 2009; Cahn et al., 2010; Krygier et al., 2013; Kakumanu et al., 2018; R.J. 2019; Burke, 2012; Martínez Vivot et al., 2020). This implies that the meditators' attentional scope shifts among various (bodily) objects of attention over time. Importantly, practitioners in this meditation maintain a narrow focus on multiple bodily sensations, moving their attention across the body rather than consistently tethering it to a single stimulus as in Mantra meditation (Nash et al., 2013).

In the present study, we aimed to investigate whether the width of attentional focus and the variation in the number of selected objects during different forms of meditation were related to distinct neural

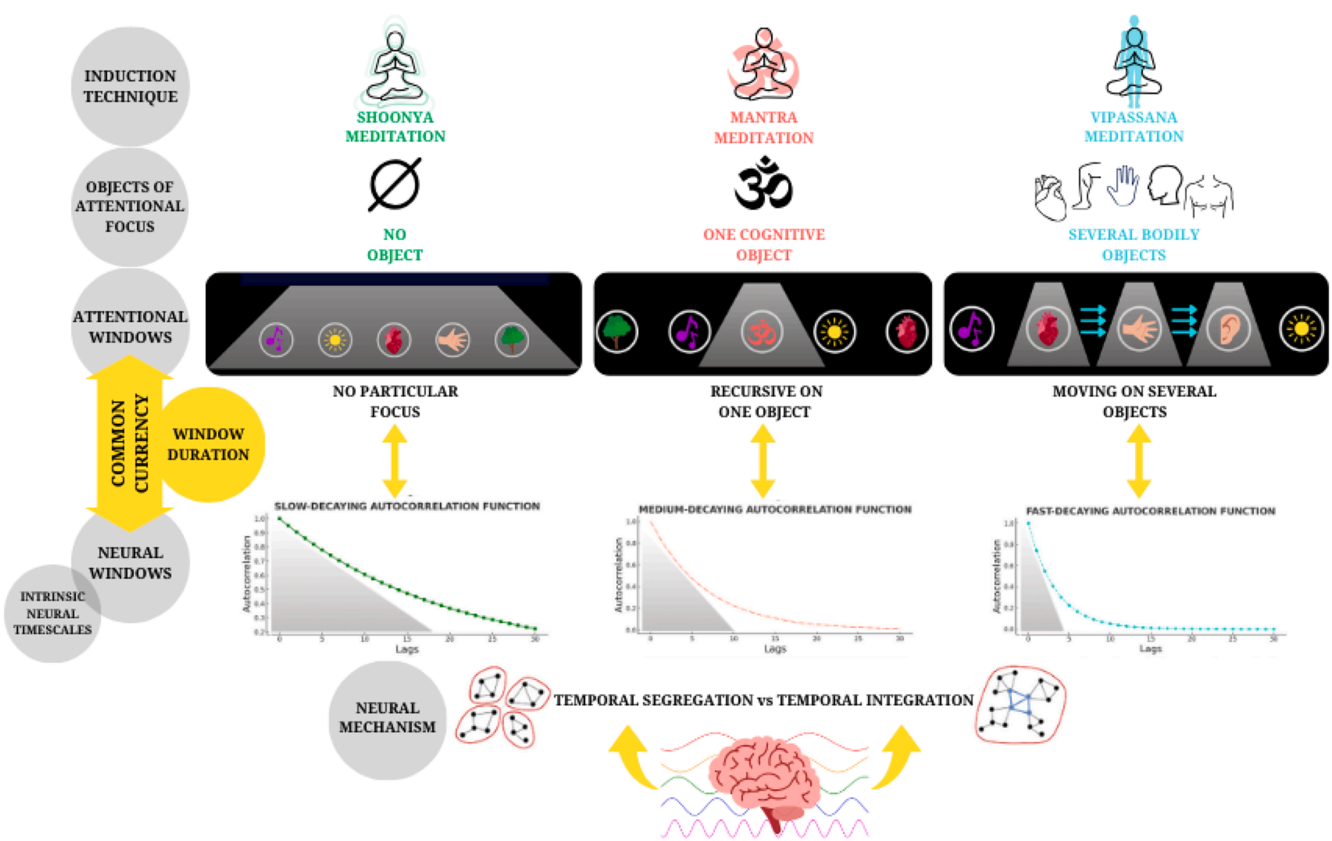

Fig. 1a. The meditation practices included in our study are characterized by a chosen object of attentional focus and by a particular bandwidth of attention. We characterized the attentional windows of attention as a spotlight: in Shoonya meditation, the window of attention spans across all inputs, without any active engagement. Conversely, in Mantra meditation, the window of attention (and thus the spotlight) is narrow and fixed only on one mantra, neglecting all other inputs in the attentional field. Finally, in Vipassana meditation, the window of attention is narrow but moving among several bodily objects. We here illustrated the hypothesized temporal link between neural windows of activity in the brain and the width of the attentional windows employed in the different meditation techniques, i.e., common currency.

substrates. Accordingly, focusing one's attention widely or narrowly on different objects means that practitioners employ different attentional windows, most likely corresponding to distinct neural temporal windows (Northoff et al., 2020 a and b). For instance, a wider attentional focus, as in Shoonya meditation, may be mediated by longer temporal windows, while a narrow attentional focus on specific stimuli, as in Mantra and Vipassana meditation, may require shorter temporal windows in the brain's neural activity. In this context, we intended to investigate if the range of attentional windows associated with these meditation techniques was mediated by a corresponding temporal window duration in the brain, that is, its intrinsic neural timescales (INTs - Golesorkhi et al., 2021a,b; Wolff et al., 2021, 2022).

1.2. Phenomenological and neural evidence of different attentional windows in various meditation traditions

The combination of phenomenology and past neural findings for each of discussed meditation techniques can shed light on the relationship between the attentional window sizes and INTs. As introduced in the previous paragraph, Shoonya meditation encourages a wide attentional window which aims at encompassing all internal and external stimuli without any active engagement (Braboszcz et al., 2017, Martínez Vivot et al., 2020; Nash et al., 2013; Laukkonen and Slagter, 2021). Previous studies showed that this and other analogous forms of open attention lead to heightened power in the slow frequency range, particularly in the theta band (Braboszcz et al., 2017; Travis et al., 2010; Baijal and Srinivasan, 2010; Berman and Stevens, 2015). An increase of power in slow frequencies has been shown to correspond to a lengthening of temporal windows in the brain (Blackman and Tukey, 1958; Kay, 1988; Zilio et al., 2021; Gao et al., 2020). In accordance, we hypothesized that this broadening of attention in such content-unspecific types of meditation may reflect wider attentional windows and, correspondingly, longer temporal windows in the brain (i.e., longer ACW).

In contrast, Mantra meditation requires practitioners to concentrate on a repeated mantra, effectively narrowing their attentional focus and excluding all the other contents present in one's attentional landscape (Braboszcz et al., 2017; Fox et al., 2016; Álvarez-Pérez et al., 2022; Tseng et al., 2022; Sharma et al., 2022; Harne et al., 2018; Nash et al., 2013; Martínez Vivot et al., 2020; Lippelt et al., 2014; Raffone and Srinivasan, 2010). A selective and narrow attentional focus has previously been shown to increase power within higher frequencies, particularly within the gamma band (Ray et al., 2008; Golesorkhi et al., 2021a). In line with this, previous studies have shown that focused types of meditation practices are associated with high power in gamma bandwidth (Braboszcz et al., 2017; Vivot et al., 2020; Travis and Shear, 2010). We thus proposed that Mantra meditations' narrowing of the attentional focus might be inextricably tied with shorter temporal windows in the brain (i.e., shorter ACW; Blackman and Tukey, 1958; Kay, 1988; Zilio et al., 2021; Gao et al., 2020).

Similarly, Vipassana meditation involves a narrowing of attention, but the focus shifts as practitioners direct their attention to various bodily sensations. The attentional windows remain relatively narrow, with practitioners concentrating on specific body parts until they become fully immersed in the inherent sensations, before shifting their focus to another body part (Braboszcz et al., 2017; Fox et al., 2016; Chiesa et al., 2010; Delgado-Pastor et al., 2013; Nash et al., 2013; Zeng et al., 2014; X. 2015; Cahn and Polich, 2009; Cahn et al., 2010; Krygier et al., 2013; Kakumanu et al., 2018; R.J. 2019; Burke, 2012; Martínez Vivot et al., 2020). Similar to Mantra meditation, this practice has been shown to result in an increase in power in the high-frequency bands (Braboszcz et al., 2017; Cahn et al., 2010; Vivot et al., 2020; Berkovich-Ohana et al., 2012), which has been associated with shorter temporal windows in the brain (Blackman and Tukey, 1958; Kay, 1988; Zilio et al., 2021; Gao et al., 2020). We thus suggested that the narrowing of attentional focus of Vipassana meditation might be linked to shorter temporal windows in the brain (i.e., shorter ACW).

Taken together, there is strong evidence for different widths of attentional windows in the different meditation traditions which may be related to distinct temporal windows in the brain's neural activity, e.g., ACW.

1.3. From attentional windows during meditation to the brain's temporal windows in its neural activity

The temporal windows of the brain's neural activity are described by the concept of INTs (Golesorkhi et al., 2021a,b; Hasson et al., 2008, Wolff et al. 2022). The INTs are "intrinsic" as they are present in the brain's resting state, though they do show task-specific modulations (Golesorkhi et al., 2021a,b). Notably, the brain's neural activity is characterized by a multitude of timescales across the cortex (Raut et al., 2020; Ito et al., 2020; Golesorkhi et al., 2021a,b). In particular, *unimodal* regions (such as primary sensory and interoceptive cortices) are mostly involved in processing short-duration stimuli through temporal *segregation* (Lerner et al., 2011; Stephens et al., 2013; Wolff et al., 2021, 2022; Ito et al. 2020; Raut et al. 2020). On the other hand, higher-order or *transmodal* regions (e.g., lateral and medial prefrontal cortex and the default-mode network/DMN) can process long-duration stimuli including sequences of different stimuli through temporal *integration* (Lerner et al., 2011; Yeshurun et al., 2021; Golesorkhi et al., 2021a,b; Murray et al., 2014; Stephens et al., 2013; Honey et al., 2012; Simony et al., 2016; Chen et al., 2017; Wolff et al., 2021, 2022). In this framework, a wider attentional focus on various stimuli or objects (as in Shoonya mediation) may be related to their temporal integration, as mediated by longer timescales in transmodal regions. Conversely, maintaining a narrow attentional focus may necessitate temporally segregating specific sensory or interoceptive stimuli from others (as in mantra or breathing meditation). This could engage the brain's shorter timescales, primarily dominating unimodal regions.

INTs can be measured by the autocorrelation window (ACW), which computes the degree to which neural activity correlates with itself across time (Spitmaan et al., 2020; Gao et al., 2020; Cavanagh and Frank, 2015; Chaudhuri et al., 2018; Murray et al., 2014; Soltani et al., 2021). INTs measurements have been applied to a plethora of mental phenomena, such as self (Wolff et al., 2019; Northoff, 2017; Kolvoort et al., 2020; Sugimura et al., 2021, Wolman et al. 2023), level of consciousness (Huang et al., 2018), disorders of consciousness (Zilio et al., 2021, 2023), sleep (Zilio et al., 2021) and psychiatric disorders (Wengler et al., 2020; Watanabe et al., 2019; Northoff et al., 2021; Lechner and Northoff, 2023). However, no studies on the modulation of INTs during meditation or in different meditation techniques with different attentional foci have been reported so far. Using EEG, we therefore investigated whether the duration of the brain's temporal windows (e.g., the INTs as calculated by the ACW) differed among distinct meditation techniques featured by the different widths in their attentional windows.

1.4. INTs, induction methods, and attentional focus: the present study's goal, aims, and hypotheses

The goal of the present study was to investigate INTs widths characterizing distinct meditation practices, as measured by ACW. This was done utilizing EEG data from three groups of meditators whose practice was characterized by different scopes of attentional focus, as well as a naïve meditation control group. All participants underwent three different sessions, i.e., (1) a breathing meditation, (2) their own meditation of expertise (tradition-specific meditation; in this session, controls performed a second session of breathing meditation), and (3) an instructed mind-wandering (Fig. 1b for a visual schema of the study design).

To examine differences in INTs in various meditation traditions, the specific aims of the present study were: (1) to test for the effect of the different types of meditation techniques (i.e., of the variable 'group') on the length of ACW. Specifically, we investigated how the subjects

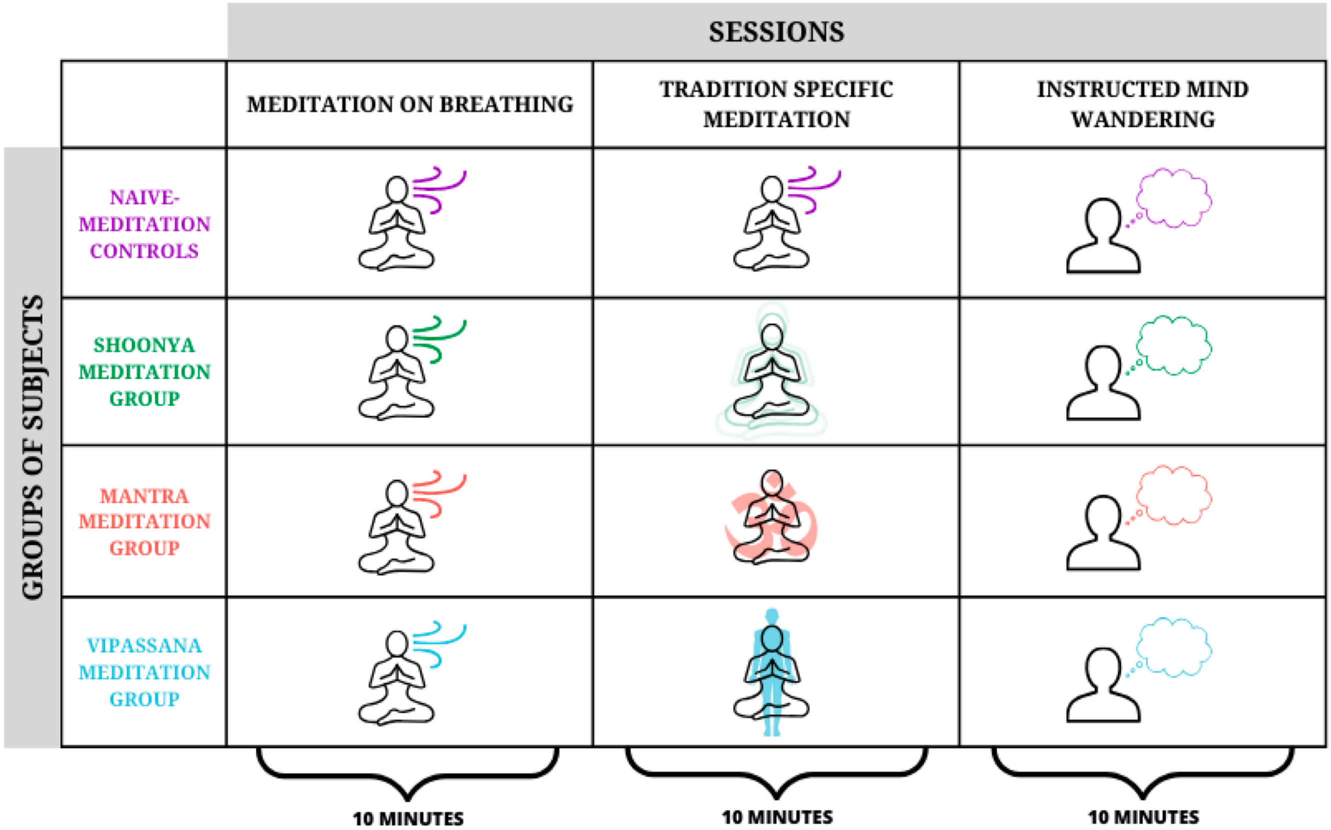

Fig. 1b. Schematic representation of the experimental paradigm of the study. The four groups of participants performed two sessions of 10-minute meditation and two sessions of 10-minute instructed mind-wandering (the findings relative to the second session of instructed mind-wandering can be found in Supplementary Material; Supplementary Figure 1). In the first meditation session, participants paid attention to their breathing. In the second one, participants had to practice their regular meditation practices, as per the meditation tradition they belonged to (during this session, the control group had to practice meditation on breathing for a second time).

belonging to different groups (Controls, Shoonya meditation, Mantra meditation, and Vipassana meditation groups) can be distinguished by different ACW lengths in their neural activity when investigating all three sessions together. The second aim (2) was to test for the effect of the sessions (breathing meditation, tradition-specific meditation, instructed mind-wandering; i.e., the variable 'session') on ACW length, investigating whether there was an ACW effect associated with each meditation type, considering together the different groups of subjects who performed them. Finally (3), we aimed to explore how the interaction between the different groups of meditators and the three sessions was characterized in terms of ACW length. We investigated this to test how ACW varied over sessions when considering the specific groups to which meditators belonged.

Our hypotheses were threefold, with respect to the three aims listed above. First, we expected to observe a significant main group effect, highlighting differences between the four different groups in their baseline INTs length as measured by ACW. In particular, we hypothesized that the Shoonya meditation group would show the longest ACW, given the widening of the scope of attention in their meditation of expertise. Second, we did not expect to find a significant main effect of the variable session alone. This is because, in the session where participants had to perform the meditation practice that was specific to their tradition (i.e., tradition-specific meditation) the four different groups were practicing different meditation types, thus possibly confounding the main effect for sessions on ACW length. Third, we expected to find a significant effect on the interaction between groups and sessions on ACW length. We hypothesized that in the session where subjects practiced their proficient meditations style, the wider attentional focus employed by Shoonya meditators would require longer temporal windows in the brain's neural activity (that is, longer ACW) thus facilitating temporal integration. Finally, we proposed that the narrow attentional focus associated with Mantra meditation as well as Vipassana meditation would require shorter temporal windows in the brain's neural activity of these two groups of meditators (that is, shorter ACW) as facilitating temporal segregation. We expected to observe these differences at the whole-cortex level, because extensive literature indicated that meditation is linked to a global neural reorganization across the whole brain rather than single regions or networks (e.g., Hasenkamp et al., 2012; Engen et al., 2017; Jao et al., 2016; Manna et al., 2010; see Fox et al., 2014, 2016; Cooper et al., 2022 for reviews on the topic). We were hence interested specifically in investigating how such global temporal reconfiguration could be characterized in terms of INTs as indexed by the ACW metric.

2. Methods

2.1. Experimental setup

We performed the analyses on an open-access dataset that can be found at the following link: *https://openneuro.org/datasets/ds003969/versions/1.0.0*. The data were recorded using a 64 + 8 channels Biosemi Active-Two amplifier system and a 10–20 Headcap standard 64-channel cap from the same company, with an original sampling rate of 1024 Hz. External electrodes, including right and left mastoid electrodes, were utilized alongside a vertical and horizontal EOG. For the EOG setup, two periocular electrodes were placed above and below the left eye, and one electrode at each of the left and right outer canthi. The procedure was conducted in a soundproof room with an electrically shielded and grounded floor. To ensure high-quality EEG signal, participants were instructed to wash their hair before the session. Additionally, the skin for non-scalp electrode placement was cleaned with an alcohol solution. All electrodes were maintained within a 50 mV offset according to the BIOSEMI system's impedance measurement metric. For further details about the dataset (data collection, procedure, ethical approval, details about the meditation traditions, etcetera) the reader is referred to Braboszcz et al. (2017).

2.2. Participants

The dataset contained data from 20 Vipassana meditators (mean age $= 47 \pm 15$), 27 Mantra meditators (mean age $= 49 \pm 13$), 20 Shoonya meditators (mean age $= 40 \pm 10$), and 32 meditation-naïve Controls (mean age $= 45 \pm 10$), for a total of 98 participants. We decided to include the control group in our analysis to more effectively assess the influence of meditation on trait characteristics and their subsequent effect on the meditative states that participants experienced. In accordance, extensive literature indicated that meditation holds a significant impact on trait features (Cahn and Polich, 2006; Harrison et al., 2018; Kiken et al., 2015; Lim et al., 2018; Tang and Tang, 2017; Bauer et al., 2019; Takahashi et al., 2005; Cooper et al., 2022 for a review). These traits represent one's own baseline that, once altered by regular meditation practice, may influence how a person enters and experiences different states of meditation (Steyer et al., 1999; R. 2015; Lee et al., 2023). Including a naive-meditation group allowed to control for these factors. The groups' estimated hours of lifetime meditation experience are summarized in Table 1.

2.3. Data cleaning

After the automatic preprocessing (see *Pre-processing* section for a detailed explanation) the remaining sessions were visually inspected and the ones that contained artifacts were manually rejected. Then, subjects with 2 or fewer sessions were deleted. Since the remaining missing data in our study, which amounted to about 3 %, were missing completely at random, i.e., independently of both observed and unobserved data, we used Expectation Maximization to impute missing data for subjects lacking only one session. This process enabled us to reach a total sample size of 85. Expectation Maximization is a powerful iterative statistical algorithm for obtaining maximum likelihood estimates of parameters when there is missing data. Further, we tested for normality in our data distribution, and after visual inspection and observation that the skew and kurtosis values were higher than ± 1.5, we employed a log transformation to fit a normal distribution. Finally, to clean the data, we winsorized it at $z = \pm 2.58$, given our sample size of 85, as per statistical guidelines. The participants thus resulted in 15 Vipassana meditators, 24 Mantra meditators, 19 Shoonya meditators, and 27 meditation-naïve Controls.

2.4. Ethics statement

Informed written consent before participation was obtained from all participants. The project was approved by the local MRI Indian ethical committee and the ethical committee of the University of California San Diego (IRB project #090,731).

Table 1
Groups estimated hours of lifetime meditation experience.

	Lifetime Hours of Meditation			
	Controls	Shoonya Meditators	Mantra Meditators	Vipassana Meditators
Mean	0.000	3259.653	14,309.167	4982.685
Std. Deviation	0.000	2866.395	14,361.270	3616.405
Minimum	0.000	760.417	1095.000	1368.750
Maximum	0.000	10,402.500	62,415.000	12,318.750

2.5. Procedure

Participants performed two sessions of 10-minute meditation and two sessions of 10-minute instructed mind-wandering (we reported here the findings relative to the first session of instructed mind-wandering; results relative to the second session can be found in Supplementary Material; Supplementary Figure 1). All sessions were performed with eyes closed. In the first meditation session (breathing meditation) participants paid attention to their breathing. In the second session (tradition-specific meditation) participants practiced their regular meditation practices, as per the meditation tradition they belonged to. During this session, the Shoonya meditation group practiced Shoonya meditation, letting go of any specific attentional focus; the Mantra meditation group practiced Mantra meditation, mentally repeating a mantra; and the Vipassana group practiced Vipassana meditation, focusing on a sequence of bodily stimuli. In this session, the control group had to practice breathing meditation for a second time. Considering previous research by Cahn et al. (2009, 2010), an instructed mind-wandering session was used as a control mental state, wherein participants were instructed to consistently recall non-emotional autobiographical memories. This method was chosen over mere resting state instructions to prevent meditators from unintentionally entering a meditative state while at rest, as commonly reported by many practitioners (Fig. 1b). The order of the sessions was counterbalanced between participants to prevent any order effects.

2.6. Pre-processing

The EEG data preprocessing was performed using the EEGLAB toolbox for MATLAB (R2023a; Delorme and Makeig, 2004). First, we cut all the 10 min sessions down to 9.5 min to avoid artifacts due to the start and the end of the EEG recording. As the original sampling rate was 1024 Hz, we down-sampled it to 512 Hz using EEGLAB's resample function. The continuous data were then band-pass filtered from 0.3 to 45 Hz. Further, the *clean_artifacts* EEGLAB plugin was applied to remove flatline channels, low-frequency drifts, noisy channels, and short-time bursts from each EEG channel. Next, the recordings were referenced to the average channels' activity. Finally, stationary artifacts, specifically eye movements, breathing artifacts, and muscular noise, were reduced using ICA and MARA EEGLAB plug-ins.

2.7. Temporal analysis

The autocorrelation function measures the similarity of a timeseries with a time-lagged version of itself by correlating the signal with itself on different time lags (Honey et al., 2012; Murray et al., 2014; Golesorkhi et al., 2021a,b). For signal x, the autocorrelation function C at lag l can be computed as:

$$C(l) = \frac{\sum_{t=1}^{N-l}(x_t - \overline{x})(x_{t+l} - \overline{x})}{\sum_{t=1}^{N}(x_t - \overline{x})(x_t - \overline{x})}$$

Where $\overline{x}$ denotes the mean of signal x and N is the number of sampling points.

The ACW was computed on the preprocessed signal. For that purpose, custom scripts were developed to compute the ACW-0 by measuring the first lag l where the autocorrelation function (ACF) reaches zero ($C(l) = 0$) which quantifies the time it takes for a signal to become decorrelated from itself (Honey et al., 2012; Murray et al., 2014; Golesorkhi, 2021a, Wolman et al., 2023; Smith et al., 2022). Consequently, the unit of the ACW-0 was seconds. In the main text, we presented the ACW values performed on the whole 9.5 min of recording. To control for stationarity during the 9.5-minute analysis interval, we adopted the methodology used by Honey et al. (2012). This approach involved computing the ACW using 20-second sliding windows with a 50 % overlap, which facilitated the assessment of data stability over time. We visually inspected these windowed ACW measurements and verified their stationarity. A graphical representation has been provided in the Supplementary Material (Fig. 2). Moreover, all the analyses were performed in these windowed ACW calculations again, and we presented the confirmatory results in the Supplementary Material. All autocorrelations were calculated on the 9.5-minute EEG data of the four sessions, for each channel, before being averaged across all electrodes to extract one ACW value per participant for each session.

We decided to investigate differences in ACW across all electrodes to align with previous research utilizing the ACW index (Wolman et al., 2022; Zilio et al., 2021; Smith et al., 2022; Buccellato et al., 2023). Extensive literature indicated that meditation is linked to a global neural reorganization (e.g., Hasenkamp et al., 2012; Engen et al., 2017; Jao et al., 2016; Manna et al., 2010; see Fox et al., 2014, 2016; Cooper et al., 2022 for reviews on the topic), prompting our interest in investigating how different meditation techniques could be characterized by INTs as indexed by global average variations in the ACW metric. We included topographical plots to provide a qualitative visualization of the ACW index across electrodes, in order to further support whole-cortex level differences.

Importantly, we believed that the ACW index, despite its link to spectral power characteristics of the EEG signal (see Supplementary material for further analysis), offered additional valuable insights. Specifically, it helped us understand how various meditation techniques were characterized by different intrinsic neural durations, which in turn influence the width of the attentional scope. This can then allow for an increasing depth of analysis with respect to the temporal common currency between psychological and neural processes (Northoff et al., 2020 a and b; Northoff, 2024).

2.8. Statistical analysis

Comparisons of the ACW were performed via a repeated measure analysis of variance with a Huynh-Feldt correction for sphericity violations in SPSS (IBM Corp, 2020). The between-subjects factor was the four groups, i.e., Controls, the Shoonya meditation group, the Mantra meditation group, and the Vipassana meditation one; and the within-subjects factor was sessions, i.e., breathing meditation, the typical meditation of the contemplative tradition each group of meditators belonged to (tradition-specific meditation) and instructed mind-wandering.

3. Results

A repeated measure ANOVA was conducted to examine the effect of meditation groups (Controls, Shoonya meditation group, Mantra meditation group, Vipassana meditation group), sessions (breathing meditation, tradition-specific meditation, instructed mind-wandering) and their interaction on ACW values, as a proxy of INTs.

3.1. The effect of groups: the impact of the induction technique

We found that the multivariate effect for groups alone ($F(1, 81) = 2.82$; $p = .044$; $\eta_p^2 = 0.094$) was statistically significant, highlighting a significant effect of the variable 'groups' on ACW length. Post-hoc comparisons indicated statistically significant differences between the

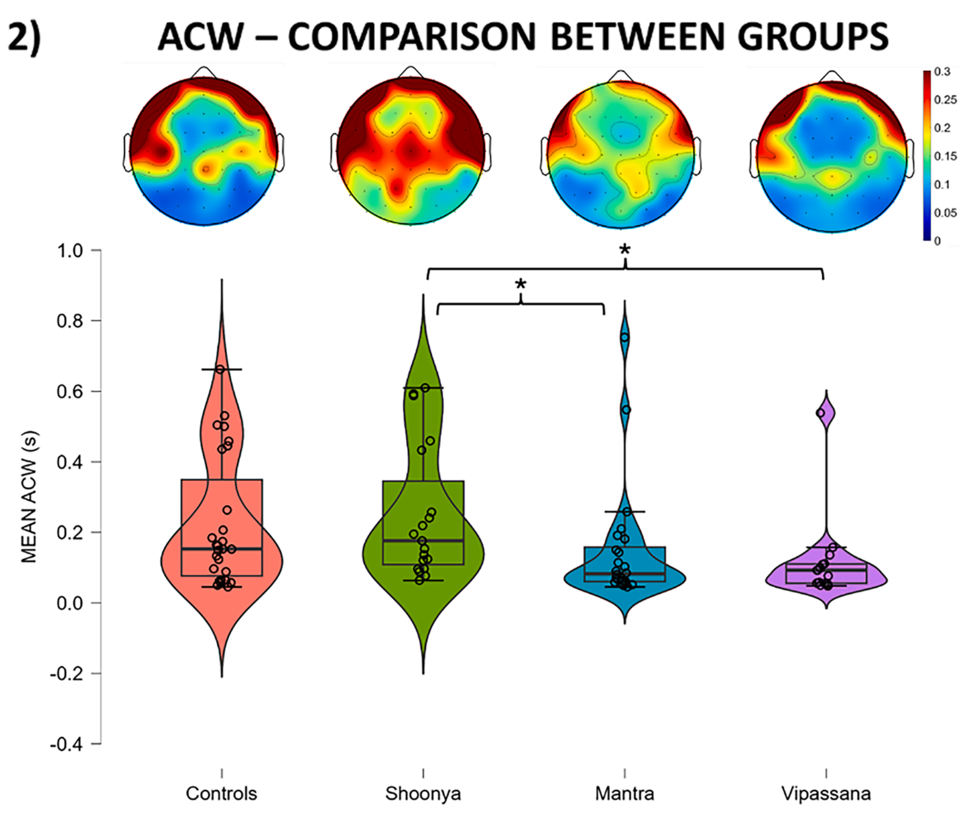

Fig. 2. The ACW distributions of the four groups (when considering the four sessions they performed together) are plotted, along with their topographical maps. Results confirmed our hypothesis that the wide focus of attention which characterizes Shoonya meditators corresponded to a longer ACW, while the narrow focus employed by Mantra and Vipassana meditators was linked to shorter ACW. The topographic maps revealed significant variations in ACW values across the entire cortex, and particularly in central, frontal, temporal, and parietal electrodes. Significance levels indicated as: * for $p \leq 0.05$, ** for $p \leq 0.01$, and *** for $p \leq 0.001$.

Mantra meditation and Shoonya meditation group (p = .034), between the Vipassana meditation and the Shoonya meditation group (p = .015), and a marginally significant difference between Controls and the Shoonya meditation group (p = .061) (Fig. 2). The topographic representations in the four groups provided qualitative support for our findings, illustrating substantial variations in ACW values across all cortical regions, and in particular in central, frontal, temporal and parietal electrodes (Fig. 2).

3.2. The effect of sessions: the impact of the different sessions

Conversely, we did not find a statistically significant multivariate effect for sessions alone (F(3, 79) = 0.704; p = .553; $\eta_p^2 = 0.026$), suggesting that when considering the four groups together there was not a statistically significant difference in ACW length between the sessions. This may indicate that the ACW remained rather stable across sessions.

3.3. The effect of the interaction between groups and sessions

We found a statistically significant effect of the interaction between sessions and groups (F(9, 243) = 3.078; p = .002; η_p^2= 0.102). This indicated that the ACW differs across sessions based on the group of meditators to which participants belong, meaning that it varied over sessions only when considering the group of which meditators were part. We thus proceeded to test for pairwise comparisons.

3.3.1. Between-group comparison

Our results showed a significantly different length of the ACW in the session tradition-specific meditation, i.e., in the session where the different groups were asked to perform the meditation specific to their tradition. We found differences between the Shoonya meditation group and the Mantra meditation group (p = .012), the Shoonya meditation group and the Vipassana meditation group (p < .001), and a marginally significant difference between the Shoonya meditation groups and Controls (p = .052). Interestingly, the Shoonya meditation group showed a longer ACW with respect to the other three groups, which is well compatible with the wider or lacking attentional focus entailed in this type of meditation (Fig. 3a).

In this same session, we also found a statistically significant difference between Controls and the Vipassana meditation group (p = .013), where the Vipassana meditation group showed a shorter ACW, possibly signifying a narrower attentional window employed during this meditation (Fig. 3a). We also observed a statistically significant difference between the Mantra meditation group and Controls (p = .046) in the session of breathing meditation (Fig. 3b). Finally, we found statistically significant differences in the session of instructed mind-wandering between the Mantra meditation group and Controls (p = .048), the Mantra meditation group and the Shoonya meditation group (p = .030), and between the Vipassana meditation group and the Shoonya meditation one (p = .049) (Fig. 3c). The topographic maps further substantiated these findings by highlighting considerable differences in ACW values across all cortical regions, and particularly illustrating major changes in central, frontal, parietal, and temporal regions (Fig. 3).

3.3.2. Within-group comparisons

Within the Shoonya meditation group, a significantly different length of the ACW values between breathing meditation and tradition-specific meditation was found (p < .001; Fig. 4a). Moreover, we observed a statistically significant difference between tradition-specific meditation and instructed mind-wandering (p = .014; Fig. 4a). In our data, tradition-specific meditation showed longer ACW with respect to the three other sessions (Fig. 4a), possibly reflecting the widening of one's attentional window during this practice, due to the lack of focus required in this practice.

Further, we found statistically significant differences in the Vipassana meditation group between the breathing meditation and tradition-specific meditation (p = .007) sessions (Fig. 4b). Specifically, the tradition-specific meditation session showed shorter ACW values compared to the other two sessions: this may be linked to the practitioners' employment of a constrained attentional window around the objects of focus and to the sequential shifting of their attentional focus on different bodily parts during the meditation session (Fig. 4b). These results were additionally supported by the topographical maps provided in Fig. 4, which illustrated the differences in the distribution of ACW values across all cortical regions. Specifically, the variations in temporal structure (as indicated by ACW measurements) were evident in frontal, central, temporal, and parietal electrodes within the Shoonya meditators group (Fig. 4a), and in temporal, parietal, and frontal electrodes within the Vipassana meditators group (Fig. 4b).

Conversely, we did not find statistically significant differences across sessions within the Mantra meditation and the Controls groups.

4. Discussion

Different meditation techniques are related to unique ways of focusing one's attention, including wider and narrow attentional windows (Braboszcz et al., 2017; Jha et al., 2007; Lutz et al., 2008; Raffone and Srinivasan, 2010; Kozhevnikov et al., 2022; Tsai and Chou, 2016; Lippelt et al., 2014; Valentine and Sweet, 1999; Manna et al., 2010; Prakash, 2021; Brandmeyer et al., 2019; Amihai and Kozhevnikov, 2015). Therefore, the present study investigated the neural mechanisms that underlie distinct meditation techniques featured by their different widths of attentional windows in the practitioners' attentional foci.

Our findings clearly showed an association between the width of one's scope of attention and the length of the INTs (as estimated by the ACW metric) observed in the practitioners' neural activity (Fig. 5). Specifically, our analysis revealed a substantial effect of the meditation group on ACW. This signified that the group to which meditators belonged influenced the brain's temporal dynamics, characterizing these practitioners by different ACW lengths in their neural activity. Specifically, the Shoonya meditation group exhibited significantly longer ACW compared to both the Mantra and Vipassana groups.

Moreover, the interaction between group and session suggested that the meditation group to which participants belonged played a crucial role in influencing the observed variations in ACW lengths across different sessions. In particular, in the tradition-specific meditation session, the Shoonya group showed significantly longer ACW compared to the other groups. This demonstrated that the wider employment of one's scope of attention is associated with longer ACW (i.e., with longer temporal windows in one's neural activity). More precisely, a prolonged ACW can be attributed to increased temporal integration of stimuli (Himberger, Chien, and Honey, 2018; Northoff et al., 2020; Golesorkhi et al., 2021a). Conversely, in the same session, the Mantra and Vipassana meditations yielded significantly shorter ACW, indicating a narrowing of attentional focus consistent with the practices' emphasis on a single mantra and on detailed awareness of bodily sensations, respectively. The strong effect of the interaction between group and session was confirmed by within-group comparisons, that further revealed differences between the sessions within the Shoonya and Vipassana groups of practitioners. Specifically, the Shoonya group was characterized by a prolongation of ACW during Shoonya meditation with respect to the other two sessions. In contrast, the Vipassana group showed a significant ACW shortening during Vipassana meditation compared to breathing meditation. Remarkably, Mantra meditators showed shorter ACW compared to Controls in both breathing meditation and instructed mind-wandering sessions, indicating a consistently shortening of INTs' length even outside their specific meditation practice. Additionally, Shoonya meditators exhibited longer ACW compared to Vipassana and Mantra meditators in the instructed mind-wandering session, suggesting that their broader attentional window persisted even when the meditators were not engaged in their traditional meditation technique. These findings suggested a potential underlying stabilization in ACW duration

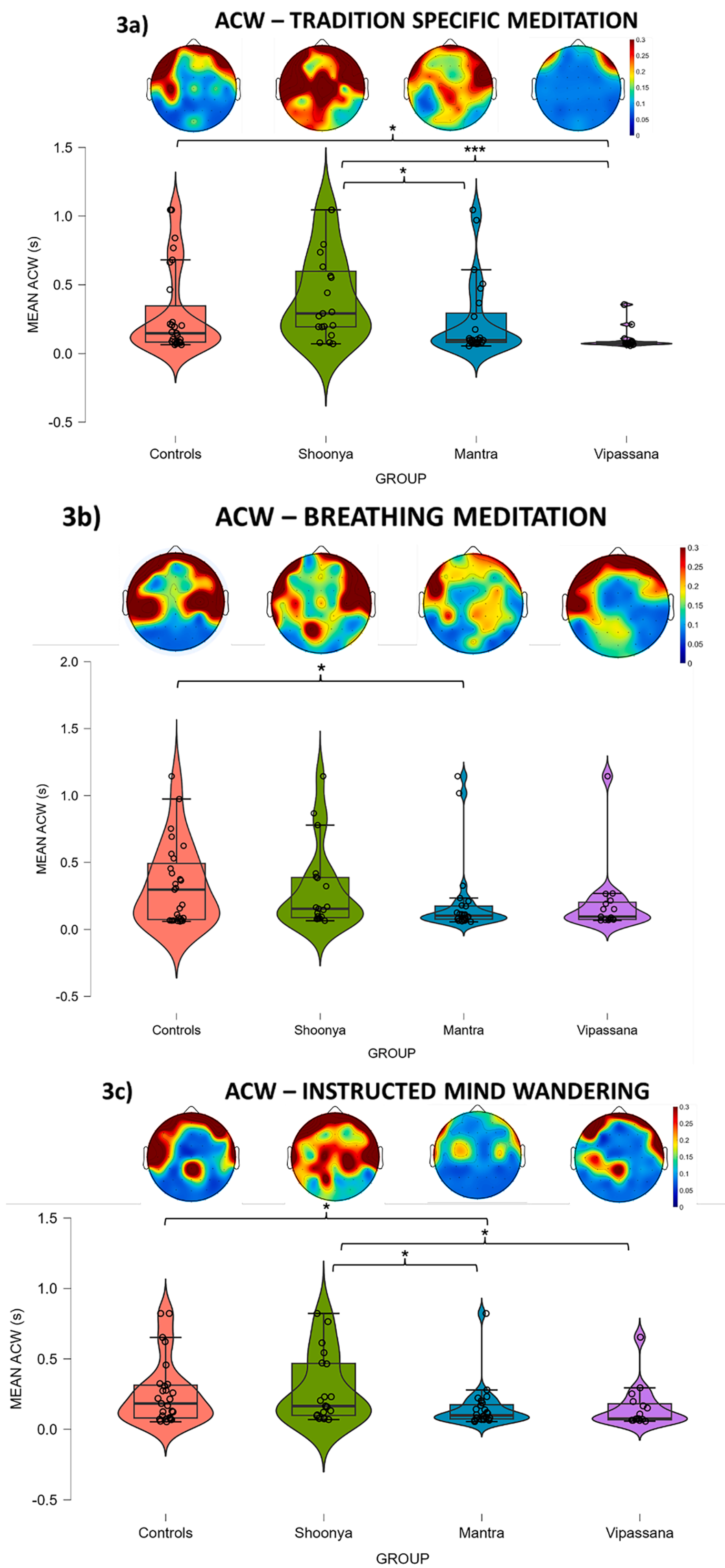

Fig. 3. The ACW distributions in (a) tradition-specific meditation, (b) breathing meditation, and (c) instructed mind-wandering sessions for the four meditation groups. Particularly, in panel a), the lengthening of the ACW for the most open kind of meditation (i.e., Shoonya meditation) compared to the ones with a narrower focus (i.e., Mantra and Vipassana meditation) is shown. The topographic maps provided support for changes in ACW across the whole cortex during these three states. Indeed, the topographic maps of panel a) illustrated major differences across all cortical regions linked to the various induction techniques employed by the four groups of participants. Here, of particular interest were the observed differences between Shoonya and Vipassana groups, which highlighted the diverse temporal structure across the whole cortex underlying different widths in the scope of attention in these meditation types. Significance levels indicated as: * for $p \leq 0.05$, ** for $p \leq 0.01$, and *** for $p \leq 0.001$.

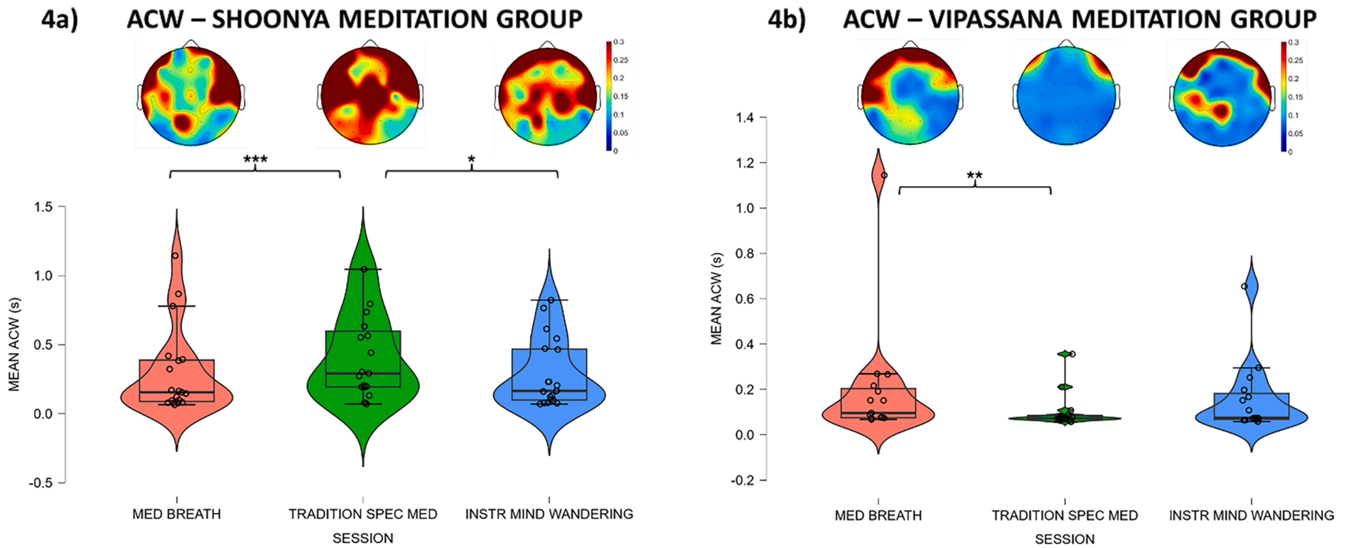

Fig. 4. The ACW distributions in the three sessions for the Shoonya meditation (a) and the Vipassana meditation (b) groups. The topographical maps illustrated that these differences are observable across all cortical regions. In particular, these variations in temporal structure as measured by ACW concern frontal, central, temporal and parietal electrodes in panel a) and temporal, parietal and frontal electrodes in panel b).
Legend: Med Breath = Breathing Meditation, Tradition Spec Med = Tradition-specific Meditation, Instr Mind Wandering = Instructed Mind-Wandering.
Significance levels indicated as: * for $p \leq 0.05$, ** for $p \leq 0.01$, and *** for $p \leq 0.001$.

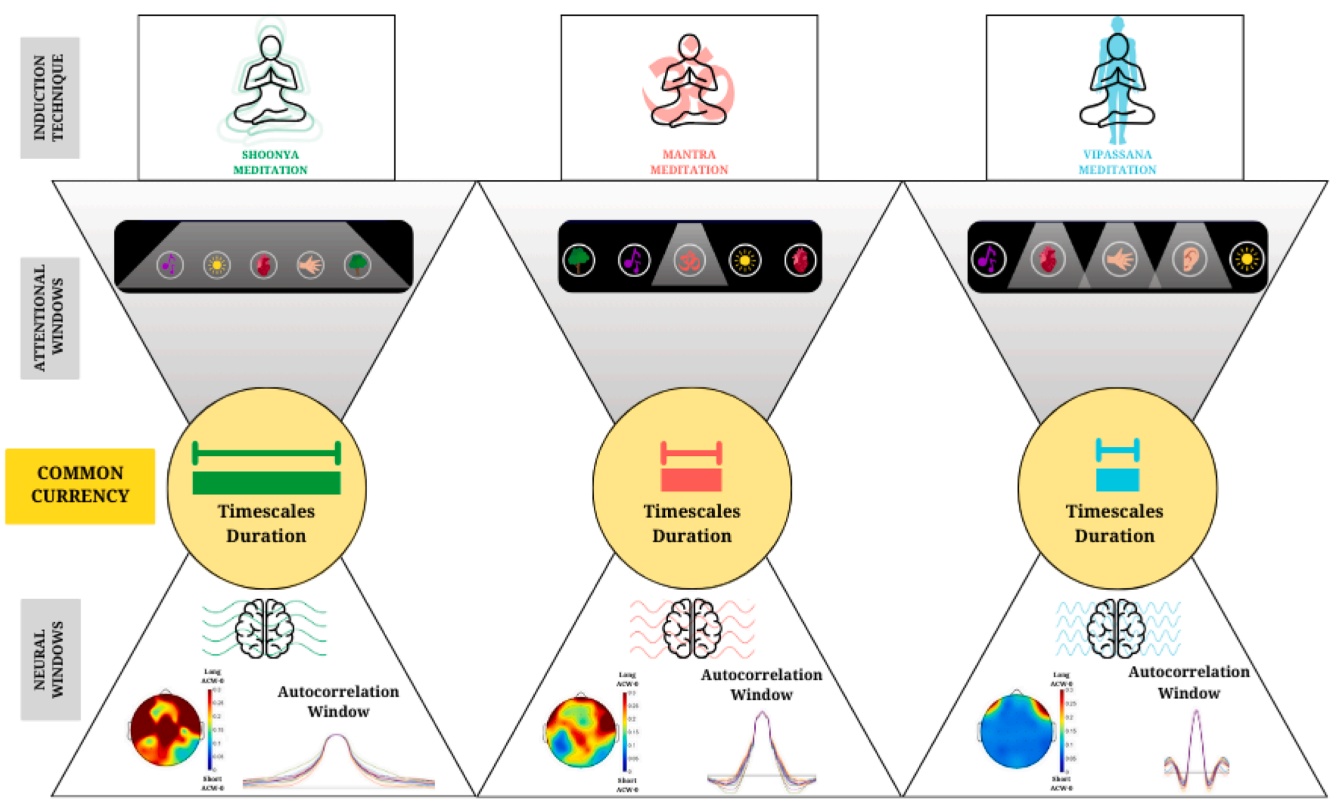

Fig. 5. The temporal relationship between the width of attentional windows and the duration of temporal windows of activity in the brain. In particular, a lack of focus such as the one that characterizes the Shoonya meditation corresponded to a longer duration INTs in the brain, which was associated with a longer ACW. Conversely, a narrower focus with shorter attentional duration such as the one that characterizes Mantra and Vipassana meditation corresponded to a shorter duration of INTs, which was associated with shorter ACW length.

differences among groups with varying meditation expertise. This stabilization was confirmed by the observed main group effect and could tentatively be linked to differences in trait features (Fig. 5).

Of note, our findings were consistent with the phenomenological reports of meditators (Berkovich-Ohana et al., 2013; Ataria et al., 2015; Nave et al., 2021; Fox et al., 2014, 2016). Indeed, open forms of meditation are characterized by a broader scope of attention that encompasses a range of objects, and a quality of attention that is often described as free-floating and unbound (Laukkonen and Slagter, 2021; Fox et al., 2014, 2016; Manna et al., 2010; Braboszcz et al., 2017, Martínez Vivot et al., 2020; Nash et al., 2013). During such practices, like Shoonya meditation, the meditator's focus of attention is less constrained to a specific object and extends to every internal and external content that arises in awareness, without any form of active engagement. Hence, this may facilitate temporal integration among the different inputs or stimuli as related to longer temporal windows on the neural level (e.g., ACW; Wolff et al., 2022).

On the other hand, the shorter neural temporal windows (as indicated by lower ACW values observed in the brain during Mantra and Vipassana meditation) may correspond to the subjective experience of a narrow attentional focus, wherein meditators deliberately exclude all other contents that may emerge in their awareness except for the selected object of meditation (Braboszcz et al., 2017; Fox et al., 2014, 2016; Álvarez-Pérez et al., 2022; Tseng et al., 2022; Sharma et al., 2022; Harne et al., 2018; Nash et al., 2013; Lippelt et al., 2014; Chiesa et al., 2010; Delgado-Pastor et al., 2013; Zeng et al., 2014; X. 2015; Cahn and Polich, 2009; Cahn et al., 2010; Krygier et al., 2013; Kakumanu et al., 2018; R.J. 2019; Burke, 2012; Martínez Vivot et al., 2020; Katyal, 2022; Lutz et al., 2008, Malinowski, 2013). Indeed, in Mantra and Vipassana meditations, practitioners have to recursively segregate the object of attention (e.g., the mantra or the sequence of body parts) from all the other contents present in one's internal and external landscape (e.g., sounds, thoughts, smells, etcetera). Interestingly, as evidenced during Mantra and Vipassana meditations, a shorter ACW (i.e., a proxy of shorter INTs) has been shown to mediate temporal segregation among inputs or stimuli in the brain's neural processing (Wolff et al., 2022). This further validated our findings and indicated a potential "common currency" bridging the experiential/phenomenological level with the neural domain (Northoff et al., 2020 a and b; Northoff, 2024).

In accordance, recent research demonstrated that the length of the brain's temporal windows in its neural activity is related to the length of the processed stimuli or inputs. For instance, a variety of studies by the group of Hasson demonstrated that shorter stimuli (i.e., single words) and longer one (i.e., sentences or even paragraphs) were related to shorter and longer timescales in the brain's neural activity, e.g., temporal receptive windows (Hasson et al., 2015; Chen et al., 2015, 2017; Yeshurun et al., 2021; Honey et al., 2012; Honey and Chen, 2022). Moreover, longer ACW in the resting state was related to higher degrees of temporal integration of temporally distant stimuli (Kolvoort et al., 2020; Wolman et al., 2023). Together, these and other studies (Wolff et al., 2022; Golesorkhi et al., 2021b) strongly suggested that the duration or length of the ACW may be related to temporal integration and segregation of different inputs or stimuli.

Following these findings, we proposed that an analogous mechanism might drive the here observed ACW differences among the distinct meditation techniques. Meditation techniques that encompass a broader attentional focus, such as Shoonya meditation, facilitated extended temporal windows. This allowed for the integration of a diverse array of inputs, stimuli, and objects, leading to longer ACW durations. In contrast, techniques with a narrow attentional focus, such as Mantra and Vipassana meditation, exhibited short temporal windows related solely to the focused stimulus, while excluding the others from the attentional landscape. This fostered temporal segregation of the different inputs or stimuli from each other, resulting in shorter ACW duration. Accordingly, we hypothesized that temporal integration and segregation of the inputs which are paid attention to provide the bridge between the brain's temporal window size and the duration of the subjects' attentional foci (Northoff et al., 2020 a and b).

Yet another mechanism related to the width of the ACW could consist in the recruitment of slower and faster frequencies. Importantly, as already mentioned in the introduction, even though previous literature on EEG correlates of diverse meditation induction techniques is far from consistent, several studies pointed out that slower frequency bands are employed in no-content meditation practices, such as in Shoonya meditation (Braboszcz et al., 2017; Travis et al., 2010; Travis and Pearson, 2000; Travis and Shear, 2010; Berman and Stevens, 2015), while faster frequency bands are employed in more content-based meditation practices like Mantra or Vipassana meditation and in similar practices (Braboszcz et al., 2017; Vivot et al., 2020; Travis and Shear, 2010; Cahn et al., 2010; Berkovich-Ohana et al., 2012). Indeed, longer ACW has been associated with increased power in the slow frequency band (Honey et al. 2012), whereas shorter ACW has been related to higher power in the fast frequency bands (Blackman and Tukey, 1958; Kay, 1988; Zilio et al., 2021; Gao et al., 2020). We thus assumed that techniques with wider attentional focus recruited slower frequencies leading to longer ACW, while those with a narrow focus rather fostered faster frequencies and subsequently shorter ACW. This relationship between ACW duration and slow fast frequency power was further supported by our finding of the negative relationship between ACW and median frequency (MF) in the whole group (see Supplementary Material).

5. Limitations, practical implications and future directions

One limitation of the current study regarded the small sample size, which may limit the generalizability of our findings. Indeed, sample size and statistical power represent a cardinal issue within the field of neuroimaging studies, with several highly influential reviews having suggested a high proportion of studies are underpowered (Button et al., 2013; Ioannidis, 2005; Marek et al., 2022). Considering that underpowered studies are prone to overestimate effect size, increase the rate of false-positives, and reduce the reproducibility of results (Button et al., 2013; Ioannidis, 2005; Marek et al., 2022), this here found link between attentional focus and INTs should be replicated to validate this interesting phenomenon.

Despite the array of meditation practices and their respective sociocultural context which may influence the generalizability of scientific results, we have shown a clear effect of the attentional window on neural activity. The practical implications of this are significant, especially as mindfulness and meditation practices are increasingly woven into mental health and well-being interventions. In this direction, understanding the neural mechanisms underlying different meditation techniques is essential. Specifically, identifying robust and replicable INT modulations of specific meditation types can help to facilitate the creation of more targeted meditation-based interventions (Lutz et al., 2014; Tang and Tang, 2020). These could have applications in stress reduction, anxiety management, and cognitive enhancement. However, developing practical and accessible means to implement these individually tailored interventions can be expensive and time-consuming. Moreover, maintaining adherence to these tailored programs can be challenging, particularly if participants do not perceive immediate benefits. Thus, formulating manageable technologies should be a priority, ensuring that they are cost-effective, user-friendly, and widely accessible, with various approaches to this beginning to show promise (Fingelkurts, Fingelkurts, and Kallio-Tamminen, 2015; Nguyen, Fdez, and Witkowski, 2024; Webb et al., 2022). This could help to overcome barriers to implementation and enhance the sustainability of these interventions.

A future direction for this research could be to include measures of the temporal structure of the stimuli/locus of attention during meditation in order to see how these may relate to brain dynamics, and to test if there is variation across individual with distinct meditational backgrounds. Another future direction could be to employ a probe-sample method while participants are meditating, in order to assess both objective (through EEG data) and subjective (through visual analogue scales' ratings) dimensions related to meditation practice (for an overview see Lutz et al., 2024). Furthermore, it would be interesting to leverage a longitudinal experimental design to underscore the long-term effects of meditation practice on ACW length. Indeed, by investigating the effect of repeated meditation sessions on ACW length modifications, we could explore both inter-individual and intra-individual changes in attentional and neural temporal windows over time. Importantly, this would allow for a richer and more fine-grained insight into the progression of attentional and neural configurations across beginner, intermediate and advanced stages of meditation training. Additionally, it would also emphasize the differences that may arise both within and between personal trajectories.

6. Conclusion

We here demonstrated that different ranges of attentional windows across meditation techniques exhibited distinct impacts onto the length of the brain's INTs. Techniques with wider foci of attention, such as Shoonya meditation in the present study, yielded larger ACW windows which we intuit as providing a wider attentional window to facilitate temporal integration of stimuli. Conversely, more attentionally focused forms of induction, such as Mantra and Vipassana meditation in the present study, showed shorter INTs, thus possibly fostering a narrower attentional window by facilitating temporal segregation of specific inputs or stimuli (from others) over integration (Wolff et al., 2022).

Together, albeit tentatively, our findings suggested a correspondence between attentional windows and the brain's neural window timescales across various forms of meditation. These temporal windows thus served to be a "common currency" (Northoff et al. 2020 a and b; Northoff, 2024) between the objective (brain) and subjective (experience) domains. Finally, our findings carried an important methodological implication when it comes to disambiguating the meditation induction techniques from the proper meditation state (Cooper et al., 2022; Reddy and Roy, 2018; J.S.K. 2019; Osho, 2012). We may need to consider the effects and neural changes associated with the induction technique itself in order to understand and clarify whether unique mental trajectories can lead to a common state of deep meditation, and how they do this. Hence, future studies may be oriented toward finding ways to differentiate between the effects of the induction technique on the one hand and the proper meditative state on the other, in order to clarify what are the neural mechanisms at play in one and the other.

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CRediT authorship contribution statement

Bianca Ventura: Writing – original draft, Formal analysis, Data curation, Conceptualization. **Yasir Çatal:** Writing – review & editing, Software, Resources, Methodology. **Angelika Wolman:** Writing – review & editing. **Andrea Buccellato:** Writing – review & editing. **Austin Clinton Cooper:** Writing – review & editing. **Georg Northoff:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization.

Declaration of competing interest

None.

Data availability

The data that support the findings of this study are openly available in OpenNeuro at doi:10.18112/openneuro.ds003969.v1.0.0, reference number ds003969. MATLAB (R2023a release) was used for this study. Most of the data analysis was conducted using the EEGLAB (http://sccn.ucsd.edu/eeglab/) toolbox, an open-source MATLAB package. Custom MATLAB scripts used in this study are available upon reasonable request. Relevant code to replicate our analysis is freely available at http://www.georgnorthoff.com/code.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.neuroimage.2024.120745.

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